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Hysteresis Phenomenon in Traffic Safety Analysis: Identification, Quantification, and Evaluation

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Introduction

Traffic Hysteresis: delay recovery from disturbances in traffic characteristics

- Scales: microscopic vehicle pair, mesoscopic platoon, macroscopic network.
- Manifest: Clockwise (CW) and Counterclockwise (CCW) loops.

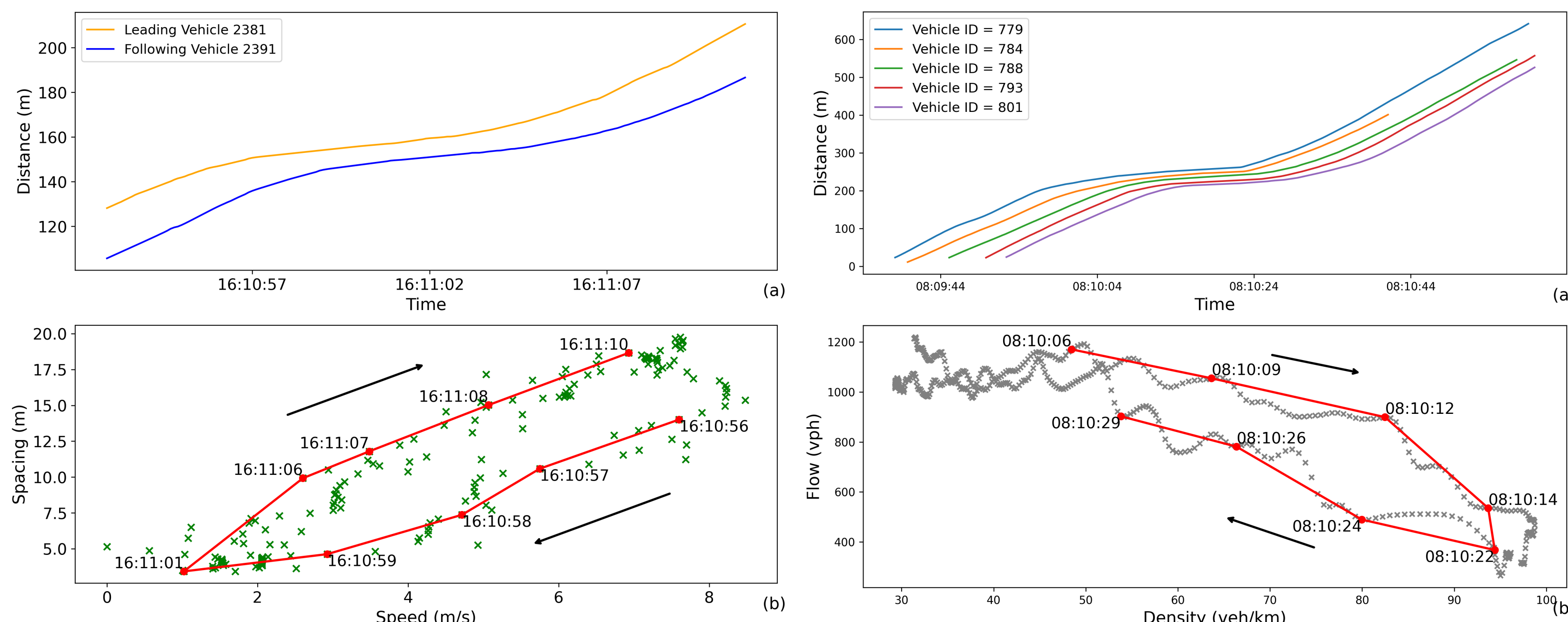


Fig.1. Traffic hysteresis at platoon level

Fig.2. Traffic hysteresis at vehicle pair level

Traffic Safety Evaluation:

- Indicators: temporal proximity, distance-based, deceleration-based indicators.
- Time-to-collision (TTC): ratio of spacing to relative speed between a vehicle pair.

Does the hysteresis phenomenon occur in traffic safety analysis?

Identification

3 Datasets: NGSIM, OpenACC, Waymo datasets

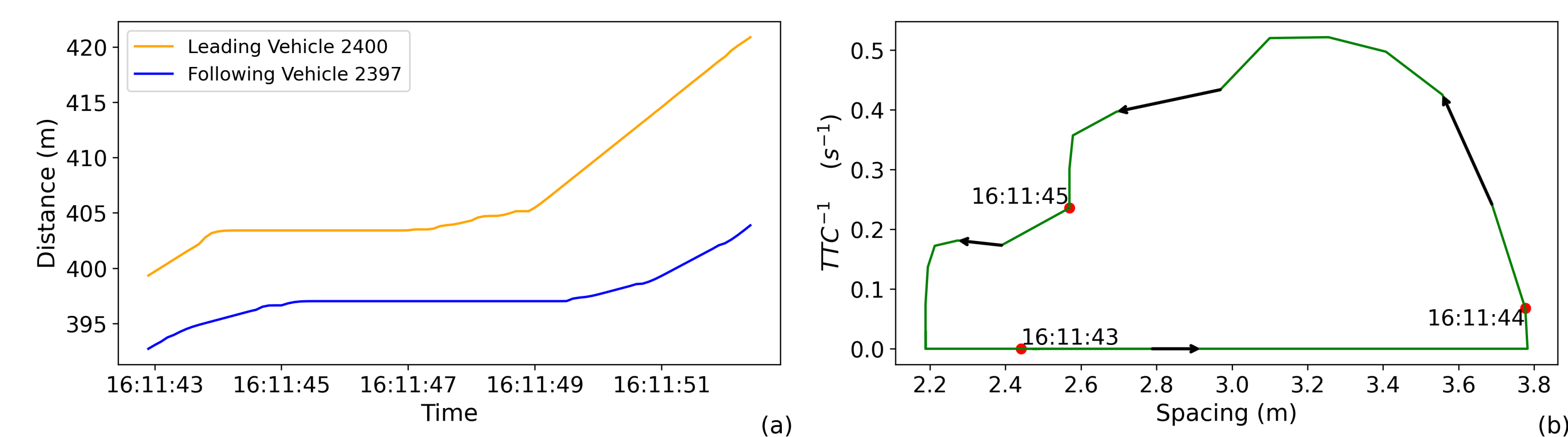


Fig.3. Hysteresis phenomenon in safety analysis for vehicle pair

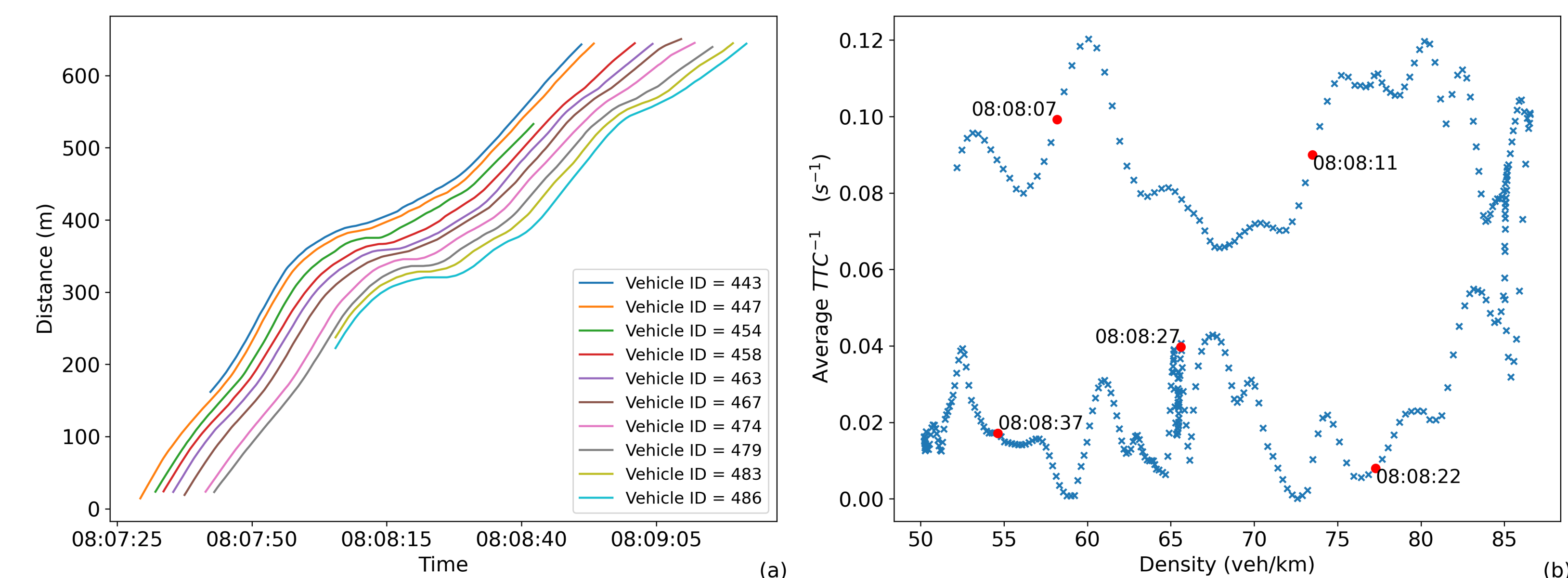


Fig.4. Hysteresis phenomenon in safety analysis for platoon

Quantification

Safety Hysteresis Intensity (HI):

$$I_s = \frac{|A|}{\Delta s_A} \quad (1)$$

- $|A|$: the area of the micro hysteresis loop A .
- Δs_A : the span of this hysteresis loop.
- $I_s > 0$ when the loop is **CCW** at **vehicle pair** level and **CW** at **platoon** level.
- A metric to quantify the magnitude of safety hysteresis phenomenon, reflecting process-level risk throughout a stop-and-go event, complementing instantaneous safety metrics.

Evaluation

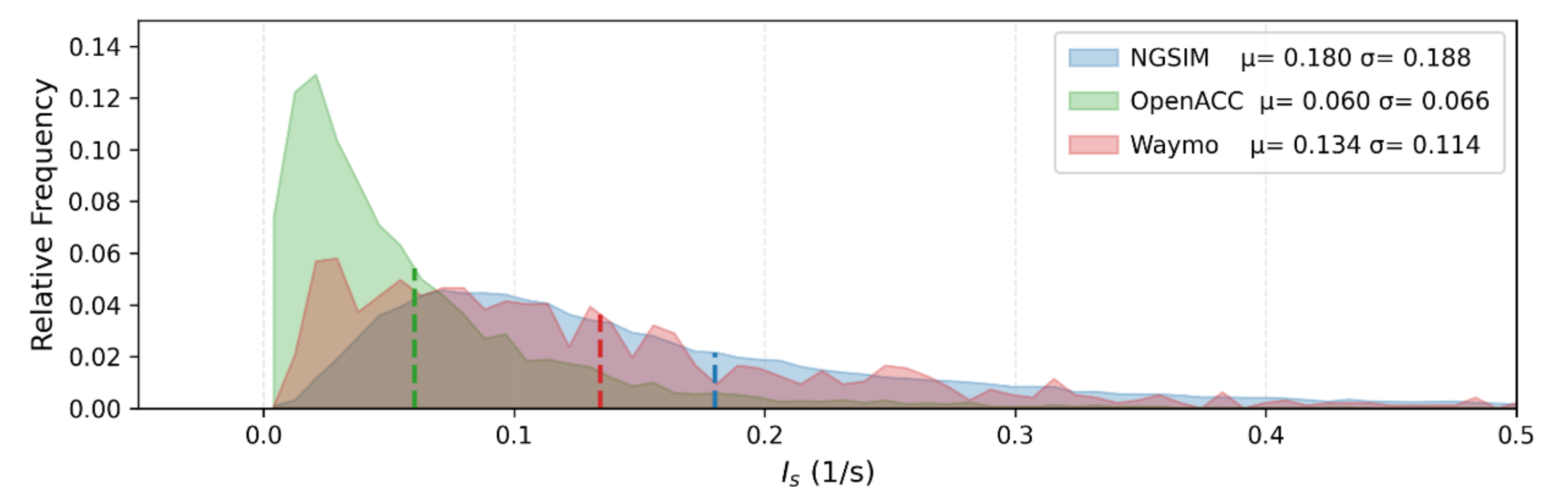


Fig.5. Distribution of positive I_s at vehicle pair level across various datasets

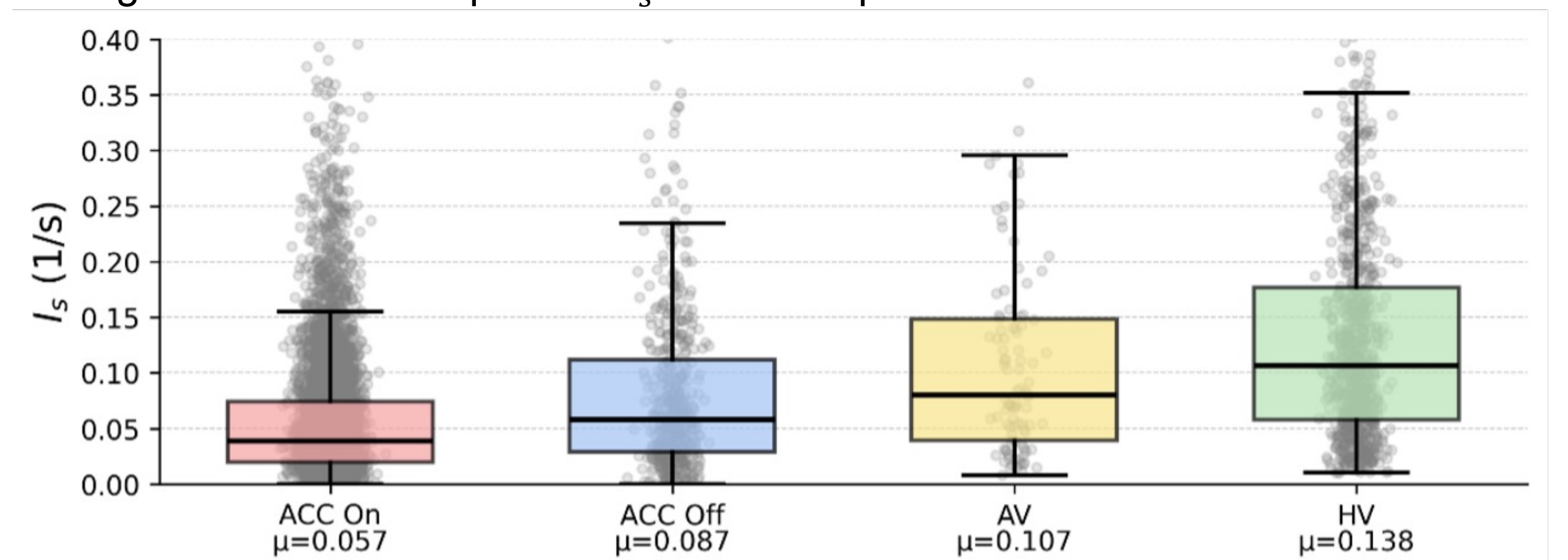


Fig.6. Distribution of I_s within vehicle pairs across different driving modes

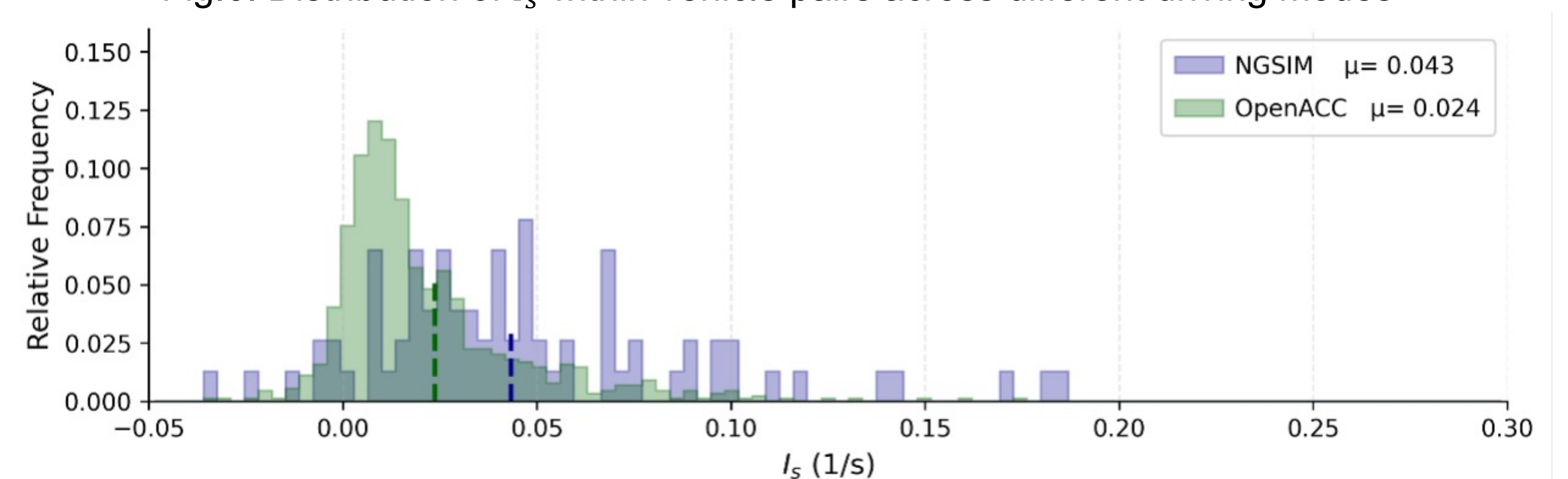


Fig.7. Distribution of I_s at platoon level across various datasets

Conclusion

Findings:

1. Safety hysteresis exists at both vehicle-pair and platoon scales.
2. Autonomous driving reduces safety HI

Contributions:

1. First identification of safety hysteresis as a path-dependent traffic safety phenomenon.
2. Safety HI: a new metric capturing process-level risk beyond instantaneous TTC.
3. Empirical basis for evaluating autonomous driving on safety hysteresis.

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